

Muntazir Mehdi (GOLD MEDALIST)

PhD Researcher at RMIT University, Melbourne

MARKING EXPERIENCE: 2 YEARS

TEACHING EXPERIENCE: 5 YEARS (Statistics)

PUBLICATIONS: 4 Research Papers

SUBMITTED: 2 Research Papers (Under Revision)

UK Graduate

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Research gate profile: <https://www.researchgate.net/profile/Muntazir-Mehdi-14>

Personal Website: <https://personal-website-blush-xi-83.vercel.app/>

An enthusiastic, experienced, hardworking, reliable, punctual, honest and determined individual with a significant teaching experience in Statistics on college and university level and currently a last year PhD researcher at RMIT University, Melbourne. A dynamic individual who is an innovative problem solver with excellent communication and timekeeping skills. I can work well on my own initiative or as part of a team and can demonstrate an ability to prioritise and work under pressure and meet tight deadlines. I am ready to accept new challenges involved in any role.

HIGHLIGHTS

- Fluent in English.
- Incorporates cross-cultural awareness.
- Strong communicator.
- Patient.
- Creative.
- Energetic.

HONORS & AWARDS

- Achieved **RMIT Tuition Fee Scholarship (RRITFS)** at RMIT University, City campus, Melbourne.
- Achieved **RTP Stipend Scholarship (RSS)** at RMIT University, City campus, Melbourne.
- Achieved **R Dennis Gibson Research Scholarship** at RMIT University.
- Won **3MT Presentation Competition (1st Position)** at RMIT University.
- Scored **Straight A grades** in the coursework of MPhil Statistics.
- Won **Chancellor's Gold Medal** in MPhil Statistics.
- Achieved more than 95% result in FBISE in my teaching career.
- Achieved 100% result in Punjab university for BCOM and ADP.

PROFESSIONAL EXPERIENCE

RMIT University
PhD Researcher

Melbourne, Australia
Feb 2024 - Present

Project: Clustering skewed datasets using newly proposed Bayesian algorithm.

Submitted Papers so far:

I have written Three papers so far in my PhD:

- 1- Developed a new Bayesian clustering algorithm for skewed, low-separation data using JAGS in R by benchmarking 30 real-world datasets against Classical and Bayesian clustering methods, improving clustering performance by up to 62% - 70% on clustering agreement and quality metrics and up to 36% on information-based measures, enabling more accurate identification of underlying population segments in skewed and visually overlapping datasets. **(Already Accepted)**
- 2- Produced operational wildfire hazard maps for New South Wales (2019--2025) by training and validating the proposed Bayesian clustering model on multi-season, high-dimensional fire data. The resulting yearly and weekly hazard maps improved wildfire early-warning performance by an average of 4% in true alarm rates and 9% in false alarm rates (7% and 15% respectively for weekly maps) compared with the next-best clustering method, helping reduce missed high-hazard events and unnecessary alerts for incident managers.
- 3- Extended the proposed Bayesian model to a fuzzy Bayesian framework for analysing uncertainty across 30 real-world datasets, integrating fuzzy set theory with probabilistic modelling. The method achieved the highest clustering performance on more than 70% of datasets evaluated and improved clustering agreement (Kappa) on 23 of the 30 datasets, while improved clustering quality (ARI), information-based measure (NMI), and the imbalanced index (IMI) on 22 of the 30 datasets compared with competing approaches, supporting more reliable clustering in the face of highly or partially imbalanced cluster sizes and balanced cluster environments.

RMIT University
Casual Faculty Staff/ Examination Invigilator

Melbourne, Australia
Aug 2024 - Present

- Assessed undergraduate and postgraduate coursework, assignments, projects, and examinations across statistics and analytics subjects, providing detailed feedback to support student learning and performance.
- Contributed to subjects including Data Wrangling, Data Visualisation, Bayesian Analysis, and Time Series Analysis, evaluating students' application of statistical methods, data analytics techniques, and interpretation of analytical results.
- Supervise undergraduate and postgraduate examinations, ensuring compliance with university assessment policies and examination procedures.
- Verify student attendance and identification, respond to examination-related queries, and maintain a secure, fair, and professional testing environment while upholding academic integrity standards.

TEACHING EXPERIENCE

Riphah International University
Visiting Lecturer

Islamabad, Pakistan
Oct 2021 – May 2022

- Delivered lectures on Statistics and probability techniques to BS students.
- Made exam papers, setup paper pattern and marked exams per the requirement.
- Build students personality by not revealing the final marks in front of class.
- Gave feedback in person rather than in front of class making them feel comfortable improving their skills.

Concordia Colleges – A Project of Beaconhouse, DHA Campus
Statistics Lecturer

Islamabad, Pakistan
Jul 2018 – Dec 2020

- Planning and delivering well-structured lessons on Statistics and mathematics topics which engage and motivate students.
- Supporting the College in delivering the curriculum effectively.
- Marked assessments, gave appropriate feedback and maintain records of pupils' progress and development.
- Stay professional and focused, manage difficult and challenging situations.

Concordia Colleges – A Project of Beaconhouse, DHA Campus
Head of Exam Department

Islamabad, Pakistan
Jul 2019 – Dec 2020

- Planning and maintaining Date-sheets.
- Supporting the College in Conducting Exam efficiently.
- Maintaining deadlines of papers/ award lists submission/ receiving.
- Planning Exam duties and seating plan of students.

PMAS-Arid Agricultural University
Visiting Lecturer

Islamabad, Pakistan
Feb 2017 – Jul 2017

- Planning and delivering well-structured lessons on Experimental Design strategies which engage and motivate students of BS.
- Supporting the Department in delivering the curriculum effectively.
- Marked assessments, gave appropriate feedback and maintain records of pupils' progress and development.
- Stay professional and focused, manage difficult and challenging situations.

Quaid-I-Azam University
Visiting Lecturer

Islamabad, Pakistan
Feb 2017 – Jul 2017

- Delivered lectures on Quality Control techniques to MSc and BS students.
- Made exam papers, setup paper pattern and marked exams per the requirement.
- Build students personality by not revealing the final marks in front of class.
- Gave feedback in person rather than in front of class making them feel comfortable improving their skills.

- Taught students on Scottish standards including High school and University level.
- Made exams and assessments which conform to standards of the school.
- Encouraged students to ask questions.

EDUCATION AND QUALIFICATIONS

RMIT University
PhD, Mathematical Sciences

Melbourne, Australia
Feb 2024 - Present

Thesis topic: Clustering skewed datasets using newly proposed Bayesian algorithm.

Riphah International University
Master of Philosophy (MPhil), Statistics

Islamabad, Pakistan
Jan 2020 – Feb 2022

Thesis topic: Bayesian Analysis of Transmuted Lomax Distribution under different loss functions.

University of Glasgow
Master of Science (MSc), Statistics

Glasgow, United Kingdom
Aug 2015 – Jul 2016

Dissertation: Inference for Item Response Data.

Quaid-I-Azam University
Master of Science (MSc), Statistics

Islamabad, Pakistan
Sep 2011 – May 2013

University of Punjab
Bachelor of Science (BSc), Maths, Statistics and Comp. Sci.

Lahore, Pakistan
Aug 2009 – Jul 2011

F.G. Sir Syed College
ICS, Maths, Statistics and Comp. Sci.

Rawalpindi, Pakistan
Sep 2007 – Jul 2009

Roots Montessori and High School
Matriculation

Rawalpindi, Pakistan
Sep 2005 – Jul 2007

PUBLICATIONS

- Mehdi, M., Demirhan, H., Taheri, S., (2026). *A clustering framework for skewed features with low true cluster separation*. Data Mining and Knowledge Discovery. - (Recently Accepted)
- Mehdi, M., Tung, Y. L., Shahzad, M. N., Khan, A. A., Akhtar, N., Alharthi, M. F. (2026). *Bayesian inference for the transmuted Rayleigh distribution: applications in behavioral data analysis*. Statistical Papers, 67(1), 5.
DOI: <https://doi.org/10.18187/pjsor.v18i4.4059>
- Hamza, M., Mehdi, M., Sardar, I. (2023). *Exploring the Prevalence and Characteristics of Child Labor in District Haripur, Pakistan: A Longitudinal Study*. Journal of Managerial Sciences, 17(2), 01–18.
URL: <https://journals.qurtuba.edu.pk/ojs/index.php/jms/article/view/727>
- Mehdi, M., Aslam, M., Feroze, N. (2022). *Bayesian Estimation of Transmuted Lomax Mixture Model with an Application to Type-I Censored Windshield Data*. Pakistan Journal of Statistics and Operation Research, 18(4), 1027–1048.
DOI: <https://www.doi.org/10.1007/s00362-026-01797-z>

CONFERENCES

- Presented research paper “*A Clustering Framework for Skewed Features with Low True Cluster Separation*” at the Australian Statistical Conference (ASC2025), Curtin University, Perth, Australia, 1–5 December 2025.

TRAININGS

- Attended 5-days ASC2025 Conference at Curtin University, Perth in December 2025.
- Attended 2-days workshop on professional development related to statistics field in Lahore conducted by Concordia Colleges.
- Attended Teacher’s Training Programme in February 2014 at Quaid-I-Azam University, Islamabad.
- Attended Orientation Programme conducted by Professor Keith Bender in University of Aberdeen, January 2015.
- Attended 5 days Math lab training conducted by Dr Ludger Evers in University of Glasgow, Glasgow.

TECHNICAL SKILLS & INTERESTS

Programming & Analytics: R (advanced), statistical modelling, Bayesian analysis, clustering, simulation, predictive modelling, data visualisation.

Data & Reporting: Data cleaning, data analysis, model validation, reporting, reproducible workflows, stakeholder presentations.

Tools: RStudio, GitHub, Microsoft Excel, LaTeX, overleaf.

Interests: Bayesian modelling | optimisation for model fitting | statistical analyses.

REFERENCES

Available on request.